

YALUTOROVSK, Russian Federation

WMO: 284650

Lat: 56.683N

Lon: 66.350E

Elev: 200

StdP: 14.59

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.7	-23.5	-34.1	0.8	-28.3	-28.6	1.1	-23.5	20.7	13.1	18.8	13.0	3.5	190	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	85.4	68.6	82.4	67.8	79.1	66.0	71.3	81.6	69.5	79.5	67.7	76.8	7.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	102.8	77.3	65.8	95.9	74.9	64.0	89.8	73.0	35.2	81.7	33.7	79.7	32.2	77.0	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	18.3	16.5		-33.8	90.8	6.1	3.2	-38.1	93.1	-41.7	95.0	-45.1	96.8	-49.5	99.1	
			DB	-33.1	73.7	6.7	2.4	-37.9	75.5	-41.8	76.9	-45.6	78.3	-50.4	80.1	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.8	1.4	5.7	21.4	39.6	53.7	62.9	65.2	61.5	50.7	37.8	20.6	7.7	
	DBStd	24.93	13.48	13.60	9.78	9.84	9.39	8.04	6.71	7.20	7.73	8.79	13.28	14.49	
	HDD50	6715	1507	1241	888	335	69	7	0	3	82	388	882	1313	
	HDD65	10876	1972	1661	1353	762	366	133	85	158	432	844	1332	1778	
	CDD50	1543	0	0	0	23	185	394	470	358	104	9	0	0	
	CDD65	228	0	0	0	0	17	69	90	49	3	0	0	0	
	CDH74	2168	0	0	0	8	203	708	798	418	33	0	0	0	
	CDH80	542	0	0	0	0	47	202	191	100	2	0	0	0	
Wind	WSAvg	7.5	7.2	7.5	8.6	8.7	8.5	7.2	6.2	6.2	6.9	7.6	7.5	7.4	
Precipitation	PrecAvg	17.6	0.8	0.6	0.7	0.7	1.6	2.5	3.0	2.1	1.9	1.4	1.3	1.0	
	PrecMax	24.4	1.6	1.2	1.7	1.8	4.0	6.9	5.6	3.5	3.7	3.6	2.7	2.1	
	PrecMin	12.3	0.0	0.0	0.0	0.0	0.2	0.5	0.7	0.9	0.4	0.0	0.3	0.4	
	PrecStd	2.6	0.4	0.3	0.4	0.5	0.9	1.3	1.3	0.8	0.8	0.9	0.6	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	37.0	46.0	74.1	85.3	88.6	88.7	87.1	78.1	65.3	45.9	36.3	
		MCWB	31.9	34.2	39.6	55.4	61.5	68.5	71.1	69.3	62.8	52.7	42.0	33.7	
	2%	DB	29.4	33.2	40.4	66.0	79.2	85.3	84.7	82.6	72.7	59.1	42.3	33.8	
		MCWB	27.9	31.2	35.7	52.1	60.4	68.0	70.3	68.6	60.5	49.9	40.2	32.0	
	5%	DB	26.1	29.7	37.3	60.0	74.8	81.9	81.7	78.4	67.5	54.3	38.6	30.0	
		MCWB	24.9	28.0	33.7	49.2	59.1	66.8	69.0	66.9	57.6	47.3	36.7	28.8	
	10%	DB	20.6	25.3	34.8	55.3	70.5	77.8	78.6	74.8	63.5	50.5	35.7	26.2	
		MCWB	19.6	23.8	32.1	46.6	56.8	65.2	67.3	65.2	55.5	45.3	33.9	24.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.2	34.9	40.5	56.9	66.5	73.3	74.5	72.3	65.2	54.4	43.0	34.0
			MCDB	33.2	36.5	44.5	69.7	77.0	83.7	83.2	82.2	75.0	62.2	44.7	35.5
2%		WB	28.1	31.3	36.0	53.1	63.0	70.1	71.8	70.0	61.3	50.9	39.8	32.0	
		MCDB	29.2	33.1	39.3	63.1	74.9	81.5	81.8	79.5	70.2	58.0	41.9	33.6	
5%		WB	24.9	28.2	34.2	50.1	60.4	68.2	70.1	67.8	58.6	48.1	36.9	28.8	
		MCDB	26.2	29.7	36.8	59.9	72.1	78.9	80.0	76.7	65.8	53.2	38.6	29.9	
10%		WB	19.7	23.8	32.3	46.9	57.8	66.2	68.4	65.8	56.3	45.5	34.1	25.0	
		MCDB	20.6	25.2	34.8	54.6	68.7	76.0	77.1	73.7	62.3	50.2	35.4	26.1	
Mean Daily Temperature Range		5% DB	MDBR	13.7	16.5	16.5	16.9	19.7	19.1	18.2	18.2	16.3	12.6	10.5	12.8
			MCDBR	13.7	14.7	14.8	24.8	25.9	25.7	24.0	24.1	22.5	20.3	10.2	11.7
	MCWBR		13.1	13.3	11.5	14.0	12.0	11.6	11.5	11.7	12.0	12.4	8.6	11.3	
	5% WB	MCDBR	13.5	14.4	13.9	22.9	23.4	22.3	22.0	21.9	20.9	19.3	9.5	11.5	
		MCWBR	13.2	13.3	11.4	13.8	12.3	11.4	12.0	11.7	12.0	12.6	8.4	11.1	
Clear-Sky Solar Irradiance	taub	0.283	0.314	0.371	0.414	0.411	0.409	0.423	0.432	0.386	0.354	0.312	0.284		
	taud	2.033	2.027	1.944	2.087	2.223	2.299	2.256	2.198	2.287	2.240	2.112	1.997		
	Ebn at Noon	178	224	239	251	262	265	257	244	239	211	166	140		
	Edh at Noon	20	30	43	44	41	38	39	39	31	25	18	16		
All-Sky Solar Radiation	RadAvg	243	560	1048	1419	1679	1822	1732	1349	870	440	213	147		
	RadStd	18	36	61	123	132	150	162	139	86	73	33	19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page